

Capital Allocation II

Estimation Risk, Bet Sizing, and Factor Decomposition

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Capital allocation under uncertainty

We know how to maximize the Sharpe ratio given known moments:

$$W^* \propto E[R^e] \text{Var}(R^e)^{-1}$$

The problem: in practice, we don't know the true moments.

- MV optimization loads heavily on assets with extreme sample means and attractive covariance properties
- We observe only *sample* estimates — the ex-post tangency portfolio, not the ex-ante one
- The portfolio that did best in-sample may be a statistical fluke

Whatever your research process, you are likely to end up quite uncertain about the alphas of your trading strategies.

Capital allocation under uncertainty

Our approach:

- ① Measure the uncertainty in our risk premia estimates for each asset
- ② Show how weights change as we perturb estimates within plausible error bands
- ③ Show how sensitive the benefits of optimization are to these perturbations

See notebook: data loading and initial MV weights.

Weights sensitivity to uncertainty

Experiment: perturb each asset's expected return by ± 2 standard errors of the sample mean, one at a time. Recompute MV weights.

The standard error of the sample mean:

$$\text{SE}(\hat{\mu}_i) = \frac{\hat{\sigma}_i}{\sqrt{T}}$$

Small changes in inputs can flip signs of portfolio weights.
Large uncertainty means your weights can be way off.

See notebook: perturbation loop over assets, weight table.

Performance sensitivity to uncertainty

Compute the Sharpe ratio for each perturbed weight vector (evaluated at sample moments):

$$SR = \frac{W' \hat{\mu}}{\sqrt{W' \hat{\Sigma} W}} \times \sqrt{12}$$

- This only perturbs *one* mean at a time
- In practice, you are also off on correlations, variances, and *several* means simultaneously
- With a long sample and constant-mean assumption, standard errors understate true uncertainty

The real test: **out-of-sample performance** (coming in Performance Evaluation lecture).

See notebook: SR sensitivity table.

Bet sizing under uncertainty

MV optimization is sensitive to inputs $\Rightarrow$ the industry developed many **ad-hoc bet-sizing rules**.

Bet sizing is one of the great skills in portfolio management — it requires instinct for uncertainty.

All rules share the structure: compute relative weights W_i , then scale by w to hit a volatility target.

Best applied after hedging common factors (market, size, value, etc.) — i.e., to residual “alpha” strategies where α_i is the expected excess return.

Mean-Variance rule

$$W_i = w \alpha_i [\Sigma_{\varepsilon}^{-1}]_i$$

If strategies are uncorrelated:

$$W_i = w \frac{\alpha_i}{\sigma_{\varepsilon,i}^2}$$

- Optimal when moments are known
- In practice, highly sensitive to estimation error in α

$1/N$ rule

$$W_i = \frac{1}{N} \text{sign}(\alpha_i)$$

- Ignore the magnitude of alpha — just bet on the direction
- Good if you have reliable sign information but noisy magnitude estimates
- Surprisingly hard to beat out of sample (DeMiguel, Garlappi & Uppal, 2009)

Proportional rule

$$W_i = w \alpha_i$$

- Weight proportional to expected return
- Ignores variance information entirely
- May be relevant if you cannot take leverage (long-only constraint)

Risk Parity rule

Assume all strategies have the same appraisal ratio: $\frac{\alpha_i}{\sigma_i} = \frac{\alpha_j}{\sigma_j}$

Substituting into the MV rule:

$$W_i = w \frac{1}{\sigma_{\varepsilon,i}}$$

This equalizes the volatility contribution of each strategy:

$$W_i \sigma_{\varepsilon,i} = W_j \sigma_{\varepsilon,j}$$

- Robust when you trust variance estimates but not return estimates
- Very popular in practice (Bridgewater, AQR)

Minimum Variance rule

$$W_i = w [\Sigma_{\varepsilon}^{-1} \mathbf{1}]_i$$

If uncorrelated: $W_i = w / \sigma_{\varepsilon,i}^2$

- Assumes all alphas are the same — uses only variance information
- Solution to: $\min_W \text{Var}(W' R^e)$ s.t. $\mathbf{1}' W = 1$
- Flip strategies so all have positive alpha before applying

Variance Shrinkage rule

Shrink Σ toward a simpler target:

$$\hat{\Sigma}_{\tau} = (1 - \tau) \Sigma + \tau \bar{\sigma}^2 I$$

where $\tau \in [0, 1]$ and $\bar{\sigma}^2$ is the average idiosyncratic variance.

Then apply MV or Min-Variance with $\hat{\Sigma}_{\tau}$:

$$W_i = w \propto \hat{\Sigma}_{\tau}^{-1}[i] \quad \text{or} \quad W_i = w \mathbf{1} \hat{\Sigma}_{\tau}^{-1}[i]$$

- Shrinks all variances toward the average, all correlations toward zero
- Stabilizes the covariance matrix — fewer extreme weights
- Generally a good idea; in some sense this *is* MV with a better Σ estimator

See notebook: all six rules applied to FF6 factors, targeted at 16% annual vol.

Portfolios and Factor models

Given N assets with excess returns R , a factor model decomposes:

$$R = A + Bf + U$$

where A = alphas, B = loadings, f = factor return, U = residuals ($E[U] = 0$).

A portfolio with weights W has return:

$$r_p = W'R = W'(A + Bf + U)$$

Expected return:

$$E[r_p] = W'A + (W'B) E[f]$$

Variance decomposition

$$\text{Var}(r_p) = \underbrace{(W'B)^2 \text{Var}(f)}_{\text{systematic risk}} + \underbrace{W' \text{Var}(U) W}_{\text{idiosyncratic risk}}$$

- **Systematic share:** $(W'B)^2 \text{Var}(f) / \text{Var}(r_p)$
- Why not just use the in-sample variance of r_p directly?
⇒ Factor structure lets you estimate the covariance matrix with far fewer parameters — critical when N is large
- **Key assumption that fails:** residuals are uncorrelated across assets
⇒ In practice, use multiple factors or allow for residual correlation structure

See notebook: CAPM regressions for SMB, HML, RMW, CMA, MOM; residual covariance matrix.

The rise of “pod shops”

Traditional hedge funds: principal traders (Soros, Robertson, Tepper, Cohen, . . .)

Modern structure: **pod shops** (Citadel, Millennium)

- Ideas generated at the pod level (5–10 people per pod)
- Capital allocated by the principal; factor risk hedged centrally

Why do pod traders work for Citadel instead of running their own fund?

Pod shop math — the $\sqrt{N}$ law

N pods, each with Sharpe ratio sr and uncorrelated residuals.

Pool expected return: $E[\sum W_i^* r_i] = N \cdot sr^2$

Pool variance: $\text{Var}[\sum W_i^* r_i] = N \cdot sr^2$

$$SR_{\text{pool}} = \sqrt{N} \cdot sr_{\text{pod}}$$

- Individually: wealth grows at rate $\sigma \cdot sr$
- In the pool: each pod's effective rate is $\sqrt{N} \cdot \sigma \cdot sr$
- Finding a new uncorrelated idea is super valuable!
- Marginal value of pod $N+1$: $\sigma \cdot sr_{\text{pool}}/N$ — decreasing, but always positive

Pod shops — caveats

This calculation **understates** the value of pooling:

- Higher SR $\Rightarrow$ more vol capacity without significant loss risk
- Wealth compounds faster

But it is also **not realistic**:

- Marginal pods are likely worse (lower SR)
- Pods are likely correlated with each other (shared factor exposures)
- Capital constraints and operational costs

Key Takeaways

- **Estimation risk matters.** MV weights are highly sensitive when signals are weak or assets are collinear — always stress-test inputs.
- **Robust bet sizing.** Simple baselines ($1/N$, vol targeting, risk parity) often outperform noisy MV out of sample.
- **Shrinkage helps.** Regularizing Σ (and sometimes μ) stabilizes weights and improves out-of-sample SR.
- **Factor lens.** Decomposing risk by factor clarifies whether diversification is *true* (low common exposures) or *illusory* (correlated bets).
- **Pod shops.** Aggregating uncorrelated strategies scales SR by $\sqrt{N}$ — the key economic logic behind the multi-PM model.

Appendix

Bet sizing rules — summary

Rule	Formula (uncorrelated)	When to use
Mean-Variance	$W_i = w \alpha_i / \sigma_i^2$	Known moments
$1/N$	$W_i = \text{sign}(\alpha_i) / N$	Reliable sign, noisy magnitude
Proportional	$W_i = w \alpha_i$	No leverage, trust means
Risk Parity	$W_i = w / \sigma_i$	Trust vol, not returns
Min-Variance	$W_i = w / \sigma_i^2$	All alphas similar
Shrinkage	$W_i = w \alpha_i / \hat{\sigma}_{\tau,i}^2$	Regularize noisy Σ

References

- DeMiguel, Garlappi & Uppal (2009) — Optimal versus naive diversification
- Ledoit & Wolf (2004) — Honey, I shrunk the sample covariance matrix
- Black & Litterman (1992) — Global portfolio optimization
- Asness, Frazzini & Pedersen (2012) — Leverage aversion and risk parity